



Figure 3: Resource Manager and Node Manager

function transforms a piece of data into key-value pairs; then the keys are sorted, where a Reduce function is applied to merge the values (based on the key) into a single output.

Resource Manager and Node Manager

The Resource Manager and the Node Manager form the data computation framework. The Resource Manager is the ultimate authority that arbitrates resources among all the applications in the system. The Node Manager is the per-machine framework agent that is responsible for containers, monitoring their resource usage (CPU, memory, disk and network), and reports this data to the Resource Manager/Scheduler.

Why Hadoop?

The problem with a relational database management system (RDBMS) is that it cannot process semi-structured data. It

can only work with structured data. The RDBMS architecture with the ER model is unable to deliver fast results with vertical scalability by adding CPU or more storage. It becomes unreliable if the main server is down. On the other hand, the Hadoop system manages effectively with large-sized structured and unstructured data in different formats such as XML, JSON and text, at high fault tolerance. With clusters of many servers in horizontal scalability, Hadoop's performance is superior. It provides faster results from Big Data and unstructured data because its Hadoop architecture is based on open source.

What Hadoop can't do

Hadoop is not suitable for online transaction processing workloads where data is randomly accessed on structured data like a relational database. Also, Hadoop is not suitable for online analytical processing or decision support system workloads, where data is sequentially accessed on structured data like a relational database, to generate reports that provide business intelligence. Nor would Hadoop be optimal for structured data sets that require very nominal latency, like when a website is served up by a MySQL database in a typical LAMP stack—that's a speed requirement that Hadoop would not serve well.

Reference

[1] <https://hadoop.apache.org/>

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